A Reduplicative Compass: Navigating Ambiguous Labels in Chinese Coordination Morphology

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1. The Paradox: An Obligatory Morphological Detour

Under the Free-MERGE assumption (Chomsky, 2013, 2015), the computational system of human language (CHL) faces a fundamental challenge in licensing symmetric structures. This paper investigates a productive reduplication pattern in Sinitic languages that reveals a novel, morphologically-driven solution to this universal problem. The AABB adjectival coordination in Sinitic language presents a sharp paradox of grammatical economy.

1.1. The Empirical Puzzle in Sinitic Coordination

Across Sinitic languages, the conjunctive coordination of a specific class of adjectives requires a seemingly complex AABB reduplicated form[1]. The corresponding simple AB form is strictly ungrammatical[2]. Consider the core data contrast from Mandarin Chinese:

(1) a. **zhe-ge* *bao-mi-hua* *tian-xian* *de*
this popcorn sweet-salty DE
(Intended: 'The popcorn is sweet and salty.')

b. *zhe-ge* *bao-mi-hua* *tian-tian-xian-xian* *de*
this popcorn sweet-sweet-salty-salty DE
'The popcorn is sweet and salty.'

c. *zhe-ge* *bao-mi-hua* *xian-xian-tian-tian* *de*
this popcorn salty-salty-sweet-sweet DE
'The popcorn is salty and sweet.' (Same meaning as 1b)

The data in (1) reveals three crucial properties. First, the AABB reduplication pattern is obligatory; the morphophonologically simpler AB base in (1a) is interestingly illicit. Second, the flexible ordering shown in (1b-c) indicates that this is a productive coordinative process, not a fixed lexical compound (cf. the fixed order of an independently existing lexical compound *ma-la* 'numbing-spicy' vs. **la-ma* 'spicy-numbing'). Third, the compositional construction denotes a strict conjunctive reading (P & Q),

[1] The ABAB reduplication pattern also exists in some Sinitic languages. However, it typically serves different functions, such as conveying verbal aspect or for stylistic effect (cf. Sui & Hu, 2021; Wang, 2022), rather than the pure adjectival coordination central to this paper. It hence falls outside the scope of the present discussion.
[2] Simple AB forms can be found in highly lexicalised compounds, such as *tian-suan-jiang* ('sweet-and-sour sauce'). Yet, these are not productive coordinations but fixed lexical items, as shown by their fixed order (*suan-tian-jiang* is not equivalent) and often specialised meanings. Hence, they are distinct from the productive, freely-ordered AABB pattern under investigation.

instead of modification or a blended state. This is clearly demonstrated by examples like (2):

(2)	*ban-ma*	*gen*	*ni*	*yi-yang,*	*shen-shang*	*dou*	*hei-hei-bai-bai*	*de*
	zebra	with	you	same	body on	all	black-black white-white	DE

'Zebras, like you, all have black and white (colour) on their bodies.'

This presents a deep paradox for any theory of grammatical economy: why does the grammar forbid a simple AB form and mandate a morphologically costly AABB structure for a basic coordination operation?

1.2. A Dual-Interface Catastrophe

I argue that the ungrammaticality of the simple *AB form stems from a catastrophic failure at the most fundamental level of syntactic computation. The input to this coordination is a symmetric Pair-MERGE of two bare, acategorial roots, ⟨√A, √B⟩. This seemingly simple operation precipitates a dual-interface catastrophe and renders the structure uninterpretable at the Conceptual-Intentional (C-I) interface and unlinearisable at the Phonetic Form (PF).

First, at the C-I interface, the structure is unlabelable. Following the Labeling Algorithm (LA; Chomsky, 2013, 2015), a syntactic object formed by two symmetric, headless elements like ⟨√A, √B⟩ cannot be assigned a label. Without a label, the C-I interface is deprived of the crucial information needed to determine the relationship between the roots (e.g., coordination vs. modification), leading to a failure of semantic interpretation.

Second, at the PF, the structure is unlinearisable. The Linear Correspondence Axiom (LCA; Kayne, 1994) requires asymmetric c-command to establish linear order. Within ⟨√A, √B⟩, the two roots mutually c-command each other, creating a symmetric structure that the LCA cannot map to a determinate sequence of sounds.

I propose that the AABB pattern is an elegant solution crafted through a sophisticated division of labor. This paper will argue that the obligatory reduplication is a morphological repair at PF, which resolves the formal crisis of the syntactic input. This repair is licensed by a specialised functional head in syntax, and its resulting symmetric form, in turn, allows the C-I interface to iconically interpret the conjunctive meaning that narrow syntax failed to derive compositionally.

2. Falsifying the Phrasal Hypothesis

The empirical puzzle of AABB coordination invites an enquiry into its syntactic status. A purely syntactic analysis, termed here the Phrasal Hypothesis (PhH), would treat AABB as a standard coordination of adjectival phrases. However, the AABB construction systematically fails key diagnostics for phrasal status and forces the rejection.

2.1. The Gradability Asymmetry

A prediction of the PhH is that the AABB construction, as a putative coordination of adjectival phrases, should itself be a gradable predicate. This prediction is false. The AABB form exhibits a strict gradability asymmetry and rejects degree modification in a way that distinguishes it from other productive adjectival constructions, including other forms of reduplication. First, the entire AABB construction cannot be modified by a single degree adverb. This is decisively shown by a language-

internal contrast in Cantonese[3]. While other productive, gradable reduplicated adjectives in Cantonese casually combine with degree adverbs (4), the conjunctive AABB form strictly resists them (5).

(4) a. *coeng4zi2* *hou2* *baak6-syut3-syut3*
wall-paper very white-snow-snow
'The wallpaper is very snowy-white.' (Cantonese, State-denoting ABB)
b. *coeng4zi2* *jau2di1* *hung4-hung4-dei2*
wall-paper slightly red-red-DIM
'The wallpaper is a little reddish.' (Cantonese, Diminutive AA-*dei2*)
(5) **coeng4zi2* *hou2 / jau2di1* *laam-luk6 / laam4-laam4-luk6-luk6.*
wall-paper very / slightly blue-green / blue-blue-green-green
Intended: 'The wallpaper is very blue-and-green.' (Cantonese, Conjunctive AB/AABB)

The non-gradability is further confirmed by the inability of AABB to participate in comparative constructions, which inherently rely on a gradable scale (6).

(6) a. *ne1-bui1* *mong1gwo2* *jyu5lok6* *syun1* *gwo3* *go2-bui1*
this-CL mango yogurt sour COMP that-CL
'This cup of mango yogurt is sourer than that one.'
b. **ne1-bui1* *mong1gwo2* *jyu5lok6* *tim4-tim4-syun1-syun1* *gwo3* *go2-bui1*
this-CL mango yogurt sweet-sweet-sour-sour COMP that-CL
Intended: 'This cup of mango yogurt is sourer and sweeter than that one.'

This non-gradability establishes a crucial empirical fact: AABB appears to be complementarily distributed with overt degree morphology. This observation invites a more specific version of the PhH, which I call the DegP Hypothesis. This hypothesis attempts to explain the complementary distribution by positing that the reduplication in AABB is itself the morphological realisation of a degree head (Deg), making the entire AABB string a DegP (cf. Zhang, 2015; Wang, 2022; a.o.). Under this view, the ungrammaticality of (5) would arise from an illicit stacking of two degree heads (e.g., *hou2* + RED). Yet, while appealing, the DegP Hypothesis is falsified by further evidence.

If AABB were a coordination of two DegPs (DegP-AA & DegP-BB), each internal conjunct should be able to host its own, distinct degree modifier. This is incorrect. As (7) shows, the internal components of AABB cannot be independently modified, demonstrating they are not syntactically independent phrases.

(7) **zhe* *dao* *cai* [$_{DegP}$ Deg-*hen* aP-*tian*] & [$_{DegP}$ Deg-*youdian* aP-*xian*] de
this CLF dish [$_{DegP}$ very sweet] & [$_{DegP}$ a-little aP-salty] DE
(Intended: 'This dish is very sweet and a little salty.')

More decisively, the DegP Hypothesis fails to explain the fundamental licensing problem that motivates the AABB construction in the first place. In Sinitic, overt degree heads like *hen* are canonical licensors of adjectival predication (Grano, 2012). They can license lexicalised AB compounds (8), but they fail to license the simple coordinative *AB form that AABB replaces (9).

[3] Cantonese data are chosen here because it exhibits higher tolerance to the co-existence of degree adverbs and adjectival reduplication (Wong, 2025).

(8) Degree licensing of lexicalised AB
zhe dao cai [$_{DegP}$ Deg-*hen* aP-*hao.chi*]
this CLF dish [$_{DegP}$ very delicious]
'This dish is very delicious.
(9) Failure of degree licensing of coordinative AB
**zhe dao cai* [$_{DegP}$ Deg-*hen* aP-*tian-xian* / aP-*xian-tian*]
this CLF dish [$_{DegP}$ very sweet-salty / salty-sweet]
(Intended: 'This dish is very sweet and salty.')

The failure of *hen* in (9) is fatal for the DegP Hypothesis. If the function of reduplication were to provide a Deg head, then an overt Deg head should be able to perform the same function and license the AB form. The fact that it cannot proves that the problem with *AB is not the absence of a degree morpheme, but an inherent, deeper structural flaw in the coordination itself. The AABB pattern, therefore, cannot be an instance of degree morphology, but must be a repair for this deeper issue.

2.2. Phrasal Inertness

A second core prediction of the PhH is that the AABB string, as a syntactic phrase, should be a mobile constituent, accessible to movement operations. This prediction is also incorrect. AABB exhibits profound phrasal inertness, resisting A-bar movement and patterning not as a syntactic phrase, but rather as a consolidated morphological word.

A crucial diagnostic for phrasal status is the ability to undergo A-bar movement, such as Topicalisation or Focus Clefting. If AABB were a well-formed phrase (DegP / &P), it should be a valid target for such operations. However, as the contrasts in (10) and (11) demonstrate, AABB systematically resists both, unlike its overt syntactic counterpart, the *you...you...* construction.

(10) Topicalisation Failure
a. [*you-tian-you-suan*]*i* *de, wo zui xihuan le* t*i*
[and-sweet-and-sour]*i* DE I most like LE t*i*
'The one which is sweet and sour, I like the most.'
b. ??[*tian-tian-suan-suan*]*i* *de*, *wo zui xihuan le* t*i*
[sweet-sweet-sour-sour]*i* DE I most like LE t*i*
Intended: 'The one which is sweet and sour, I like the most.'
(11) Focus Clefting Failure
a. *wo xihuan de shi* [*you-tian-you-suan de*]
I like DE be [and-sweet-and-sour DE]
'What I like is sweet-and-sour.'
b. ??*wo xihuan de shi* [*tian-tian-suan-suan de*]
I like DE is [sweet-sweet-sour-sour DE]
Intended: 'What I like is sweet-and-sour.'

This inability to move is inexplicable under the PhH, but it is precisely expected if AABB is rather a word, not a phrase. The behavior of AABB is perfectly captured by the Lexical Integrity Principle (Bresnan & Mchombo, 1995), a foundational principle which states that words and phrases are built by different principles, and that the internal structure of words is opaque to syntactic operations. Within the DM framework adopted here, this inertness is a direct consequence of the derivational

history of the AABB form. Syntactic movement targets syntactic constituents (XPs). Since AABB, I argue, is only formed post-syntactically at the PF interface, it lacks the status of a phrasal constituent within narrow syntax and is therefore unable to perform movement operations.

2.3. The Selectional Puzzle

The most profound evidence against the PhH lies in the peculiar and strict selectional restrictions governing the AABB construction. A general syntactic rule of coordination should be blind to the fine-grained semantic content or phonological shape of the phrases it conjoins. However, the AABB mechanism is demonstrably sensitive to exactly these non-syntactic properties, revealing its nature as a specialised, interface-governed operation.

First, the AABB pattern imposes a semantic filter. It productively applies only to adjectives denoting rather direct physical properties (12a), while systematically rejecting those denoting more abstract judgments (12b).

(12) AABB Coordination (Morphological)
a. Legal (Physical/Observable Properties):
tian-tian-suan-suan (sweet-sour), *hung-hung-hei-hei* (red-black)
b. Illegal (Abstract Properties):
**hao-hao-kuai-kuai* (good-fast), **ben-ben-huai-huai* (stupid-bad)

Second, the AABB mechanism imposes a strict phonological uniformity constraint. It requires its inputs to be uniformly monosyllabic and cannot combine adjectives of different syllable counts (13).

(13) Illegal "Mixed" Coordination:
* *hei-hei-jie.shi*/ **hei-hei-jie.shi-jie.shi* (black-muscular)
**gao-gao-shuai.qi* / **gao-gao-shuai.qi-shuai.qi* (tall-handsome)

This selectivity is inexplicable under the PhH. A syntactic coordination rule is defined over syntactic categories, not their semantic or phonological content. In stark contrast, overt syntactic coordination imposes no such restriction and can freely combine adjectives (cf. (14)).

(14) *you*-A-*you*-B Coordination (Syntactic)
a. Non-restriction of lexical class:
you-tian-you-suan (sweet and sour), *you-hao-you-kuai* (good and fast)
b. Non-restriction of prosodic count:
you-shuai.qi-you-you.qian (handsome and rich), you-gwei-you-nan.ci (expensive and taste bad)

This type of constraint, which demands formal uniformity between its inputs, is again unexpected for a syntactic rule but is a classic property of a morphological template. The failure of these mixed-syllable forms argues against a simple phonological motivation (e.g., "strengthening" light monosyllables); the operation is not about repairing individual items, but about satisfying a template that requires parallel inputs.

The convergence of this evidence, sensitivity to both semantic dimensions and phonological shape, points to an inescapable conclusion: the AABB formation is governed not by the principles of syntax, but by the properties of a specialised morphological form.

3. The Proposal: An Interface-Based Repair

Having demonstrated that no purely syntactic analysis can account for the paradoxical properties of the AABB construction, I propose a solution that lies in a division of labor between syntax and its interfaces. The AABB form, I argue, is the result of a morphological repair mechanism that resolves an interpretive crisis originating in narrow syntax. This crisis is resolved through a dedicated licensing head in syntax, a formal repair at the PF interface, and an iconic interpretation at the C-I interface.

3.1. The Syntactic Derivation: Containing a Labeling Defect

The derivation begins with the fundamental choice of how to merge two roots, √SWEET and √SALTY, to derive a coordinative reading. This requires the selection of Pair-MERGE (cf. Pan & Du, 2024), resulting in a symmetric, headless set of two acategorial atoms (15).

(15) Pair-MERGE(√SWEET, √SALTY) → $\langle_{\alpha=??}$ √SWEET, √SALTY$\rangle$

As established by the LA, the resulting object α is unlabelable. Feature sharing is also unavailable here as bare roots are feature-poor primitives, lacking the necessary formal features (e.g., φ-features) to establish a relation under the standard assumption (cf. Chomsky 2015 on roots being "too weak" to label) [4]. Subsequently, a categoriser a^0 merges with α. Following the standard algorithm, this new object is successfully labeled as aP by its head, (16).

(16) Set-MERGE(a^0, $\langle_{\alpha=??}$ √SWEET, √SALTY$\rangle$) → $\{_{aP}$ a^0, $\langle_{\alpha=??}$ √SWEET, √SALTY$\rangle\}$

While a syntactically licit aP is formed, the labeling defect of its complement α is not resolved, but merely contained within the larger phrase. This contained defect shows fatal at the interfaces. The syntactic structure of α lacks the necessary hierarchical information (due to symmetry and headlessness) for the C-I interface to compute the specific semantic relation (coordination vs. modification) between √A and √B compositionally, even if α is contained within a labeled aP. The PF interface, in parallel, cannot linearise the symmetric α. The *AB form is thus ungrammatical as it corresponds to an object that is illegible to the interface components.

3.2. The Convergent Path: Licensing, Repair, and Interpretation

For the derivation to converge, a dedicated mechanism must be available in the numeration, unfolding in three distinct but cooperative stages.

3.2.1. Syntactic Licensing by F-RED

The convergent path is made possible by a functional head, F. The necessity of such a head, rather than a purely morphological process, is evidenced by the predicative function of the AABB construction. In Sinitic languages, adjectival predication is subject to a T[V] constraint, which requires a licensing functional head (e.g., Deg) to provide a [V] feature to project a TP (Grano, 2012).

[4] I assume that the LA can label an {H, ??} configuration as long as H is a well-formed head. The minimal search of the LA is local: it probes the immediate constituents and successfully identifies the unique head H.

Since the resulting AABB form can act as a standalone predicate, see (1b-c), a functional head must be present in its underlying structure to provide this predicative feature. I propose that F serves this crucial role, carrying the [V] feature. Its second formal property is that its exponent at PF is a reduplicant morpheme (RED). Under Free Merge, F-RED merges with the aP containing the unlabeled root-pair and form a legally labeled FP (17). [5]

(17) Syntactic Licensing by F-RED:
Set-MERGE(F, $\{a^0, \langle_{\alpha=??} \sqrt{SWEET}, \sqrt{SALTY}\rangle\}$) → $\{_{FP}$ F $\{_{aP}$ a^0 $\langle_{\alpha=??} \sqrt{SWEET}, \sqrt{SALTY}\rangle\}\}$

3.2.2. The PF Realisation: The Reduplicative Cascade

At the PF interface, the abstract head F must be spelled out. Its exponence is the RED morpheme, which triggers a complex morphological procedure. This process does not "repair" the aP, but rather realises its lexical contents (√A, √B) as a new, well-formed morphological object[6].

Step 1: Transfer and Lowering

The syntactic object $\{_{FP}$ F-RED aP$\}$ is transferred to the PF component. The abstract functional morpheme F-RED undergoes Lowering (Embick & Noyer, 2001; Lee-Kim, 2017) and adjoins to the categoriser a^0. The features of F then Fission, splitting into two identical bundles, each targeting one of the roots within the complement α.

(18) a. Lowering: $\{_{FP} \ldots \{_{aP}\ a^0\text{+F-RED}\ \langle_{\alpha=??} \sqrt{SWEET}, \sqrt{SALTY}\rangle\}\}$
b. Morpheme Fission: $\{_{FP} \ldots \{_{aP} \ldots \langle_{\alpha=??} \sqrt{SWEET\text{-}RED}, \sqrt{SALTY\text{-}RED}\rangle\}\}$

Step 2: Exponence by Copying and M-Word Formation.

The exponence of the RED is Local Copying. This Vocabulary Insertion process copies the phonological matrix of each targeted root, creating two distinct, phonologically stable Morphological Words (M-Words; Embick & Noyer, 2001): $[AA]_{MWd}$ and $[BB]_{MWd}$.

(19) a. Vocabulary Insertion: $\langle_{\alpha=??}$ *sweet*-RED, *salty*-RED$\rangle$
b. Morpheme Copying: $\langle_{\alpha=??}$ $[$*sweet-sweet*$]_{MWd}$, $[$*salty-salty*$]_{MWd}\rangle$

Step 3: Merger and Linearisation.

The two adjacent M-Words are then consolidated into a single complex word via Local Dislocation Merger[7]. This operation decisively bypasses the linearisation paradox. Instead of attempting to apply the LCA to the unlinearisable root-pair ⟨√A, √B⟩, the PF component replaces this symmetric syntactic object with two distinct, newly formed morphological units (cf. Hu, 2023). As these new units ($[AA]_{MWd}$ and $[BB]_{MWd}$) are identifiable and discrete, they can be concatenated into a linear string.

[5] I use the label F-RED throughout the paper as a convenient shorthand to refer to this specific functional head (F) whose phonological exponent is the reduplicant.
[6] Adopting a dynamic phase approach (e.g., Bošković 2014), I assume the entire FP containing aP and α constitutes a single spell-out domain, transferred to the interfaces simultaneously.
[7] Local Dislocation applies under Adjacency Condition. It hence blocks discontinuous outputs like A-B-A-B.

Since the original syntactic input α was symmetric, the PF has no hierarchical information to dictate the order of concatenation. The ordering is therefore free, yielding either AABB or BBAA. The LCA is thus bypassed, not violated, as the morphological component has provided a licit output where the direct syntax-to-phonology mapping would have failed.

(20) Local Dislocation Merger and Free Linearisation:
a. [*sweet-sweet*]$_{MWd}$ * [*salty-salty*]$_{MWd}$ → [*sweet-sweet-salty-salty*]$_{MWd}$
b. [*sweet-sweet*]$_{MWd}$ * [*salty-salty*]$_{MWd}$ → [*salty-salty-sweet-sweet*]$_{MWd}$

The result of this cascade is a single, well-formed morphological word, AABB, which is the final output passed to the Articulatory-Perceptual systems.

3.2.3. C-I Interpretation: Iconicity as a Last Resort

The morphological word AABB created at PF provides the key to resolving the interpretive crisis at the C-I interface. I propose that the C-I system operates under a principle of interpretive economy: it first attempts a standard compositional interpretation based on the syntactic structure. Only if this fails does it resort to a 'last resort' mechanism: interpreting the surface morphological form directly.

A crucial question arises: why does the C-I system not apply an iconic principle directly to the symmetric syntactic object ⟨√A, √B⟩ to resolve the ambiguity? The answer lies in a fundamental principle of the architectural division of labor: the iconic interpretation mechanism at C-I operates on concrete, linearised surface forms provided by PF, not on abstract, unlinearised syntactic objects. The abstract object ⟨√A, √B⟩ is a hierarchical set, not a string with a perceptible form for iconicity to apply to. It is the PF component that provides this crucial exponent by transforming the unlinearisable syntactic object into a perceptually symmetric morphological word, AABB.

This is best illustrated by contrasting AABB with asymmetric, partial reduplications (*hong-tong-tong*, "vividly red"), which are rather derived from syntactically asymmetric structures, and their meaning is thus derived compositionally (Wong, 2025). The C-I interface can therefore successfully derive their compositional meaning from the syntactic hierarchy. In this case, the 'last resort' iconic mechanism is never triggered. The AABB construction, in contrast, forces the non-compositional route. As argued, its underlying aP contains an unlabelable root-pair α, causing compositional interpretation to fail. Faced with this failure, the C-I system must resort to its backup strategy. It analyses the symmetric morphological output provided by PF and applies the following principle:

(21) Principle of Iconic Interpretation (Last Resort):
When compositional interpretation fails, the C-I system may assign meaning based on the symmetry of the morphological form: a symmetric form like AABB is assigned a coordinative meaning.

Since the AABB form is perfectly symmetric, the C-I interface assigns it a conjunctive reading (PROPERTY(A) ∧ PROPERTY(B)), which is then integrated into the predicative frame provided by F. The interpretive crisis is thus resolved by a 'last resort' mechanism that reads meaning directly from a morphologically transparent form, but only after PF has created that form, rendering the opacity of syntax irrelevant.

4. Deriving the Signature Properties of AABB

The interface-based model, where AABB is the morphological realisation of a functional head F licensing a syntactically licit but interpretively ambiguous aP, does more than simply derive the correct surface form. It provides a unified explanation for the entire unique signature of grammatical properties that falsified the PhH in Section 2. These properties are the direct and predictable consequences of the derivational history of AABB as a post-syntactic, morphological construct.

4.1. Deriving Syntactic Inertness: A Post-Syntactically Formed Word

The proposed model, in which the AABB form is a single Morphological Word forged post-syntactically, provides a direct and unified explanation for its profound syntactic inertness. The seemingly disparate facts, its failure to undergo phrasal movement and its inability to be graded, are shown to be two sides of the same coin, both stemming from its status as a word, not a phrase.

First, the immobility of the AABB construction is a direct consequence of its derivational history. Syntactic movement targets syntactic constituents (XPs). Since the AABB form only comes into existence at the PF interface, it lacks the status of a phrasal constituent within narrow syntax and is therefore invisible to movement operations like Topicalisation or Clefting. This is a classic manifestation of the Lexical Integrity Principle (Bresnan & Mchombo, 1995), which renders morphological objects rather opaque to syntactic rules.

Second, the absolute non-gradability of AABB follows from the same principle. Syntactic degree modification, licensed by Degree heads like *hen* "very", is an operation that targets phrasal projections of gradable predicates (aPs). As a morphological word, the AABB form is not a phrasal category and thus cannot be selected by a Deg head. On the other hand, this formal restriction is mirrored at the semantic level. The licensing head F that triggers the AABB formation is a type-shifting operator that transforms the gradable predicate with a semantic type <d,et> into a simple, non-gradable semantic type <e,t>, leaving no open degree slot for modification (cf. Grano, 2012).

4.2. Explaining the Selectional Puzzle

The interface-based model provides a principled explanation for the peculiar selectional restrictions of the AABB construction. These restrictions, fatal for any purely syntactic account, are predictable consequences of the interface-driven nature of the repair mechanism, particularly the constraints imposed by the Conceptual-Intentional (C-I) interface.

The first, and most profound, restriction is semantic. While AABB can productively combine adjectives even from different sensory domains, as shown in (22), it systematically rejects those denoting abstract or evaluative judgments, as in (23).[8]

(22) Legal Cross-Dimensional Coordination:
a. *bai-bai-tian-tian* (white-sweet) (Color + Taste)
b. *hei-hei-chou-chou* (black-stinky) (Color + Smell)
c. *hong-hong-la-la* (red-spicy) (Color + Taste/Sensation)

(23) Illegal Abstract Coordination:
a. **hao-hao-shuai-shuai* (good-good-handsome-handsome) (Quality + Appearance)
b. **huai-huai-qing-qing* (bad-bad-light-light) (Quality + Weight)

[8] Overt syntactic coordination via *you...you...* ("and...and..."), in contrast, is not subject to this constraint as it simply conjoins two independent, fully formed propositions.

This pattern reveals that the constraint is not about maintaining a single semantic dimension, but about the cognitive nature of the properties involved. I propose that this filter operates at the C-I interface. As argued, the conjunctive meaning of AABB is derived via an iconic interpretation of its symmetric form. This integration is only possible for properties that are directly perceivable by the senses (e.g., color, taste, smell, shape). Abstract notions like "good", "bad", "many", or "few" are not direct sensory inputs but are the result of higher-level cognitive judgments and evaluations. The C-I system cannot "package" a direct sensory property like [+handsome] with an abstract judgment like [+good] into a single, unified percept. The conceptual integration fails, blocking the interpretive procedure and rendering the output uninterpretable.

Second, the phonological uniformity constraint is captured as a formal requirement of the PF realisation rule. The PF repair is a morphological template that operates on a symmetric pair of bare roots (√). A polysyllabic adjective like *jieshi* "muscular" or *haochi* "delicious", in contrast, is not a bare root. Within the Root Merger approach (c.f., Zhang, 2015; Hu & Perry, 2017), such items are derived from the merger of two roots (e.g., √JIE + √SHI) that has already undergone categorisation to form a lexicalised aP-*jieshi*.

The AABB template, therefore, imposes a crucial constraint on its inputs: they must be acategorial primitives (√), not already-categorised phrases (aP). An attempt to input a bare root (√HEI) and a categorised phrase (aP-*jieshi*) results in a category mismatch for the morphological rule, blocking its application. This formal distinction between a bare root and a categorised word/phrase explains the strict monosyllabic constraint observed in (24).

(24) Illegal (Mixed Syllables and Categories):
a. **hei-hei-jie.shi* (black-black [1-syllable & √] and muscular [2-syllables & aP])
b. **tian-tian-hao.chi* (sweet-sweet [1-syllable & √] and delicious [2-syllables & aP])

5. Conclusion

This paper has argued that the obligatory AABB reduplication in Sinitic adjectival coordination is an elegant, interface-driven solution to the universal problem of symmetric merger. The central claim is that the AABB form is a post-syntactic morphological repair for a structure that, while syntactically well-formed, creates an interpretive crisis at the interfaces.

This analysis reveals a novel model for the division of labor within the grammar. Syntax freely generates deviant structures containing unlabelable root-pairs, whose fate is then determined by the interfaces acting as active repair systems. The PF component licenses the form by realising the structure as a symmetric morphological word (AABB), resolving the linearisation paradox. The C-I component, in turn, licenses the content by iconically interpreting this symmetric form as a conjunctive meaning, bypassing the initial labeling failure.

The findings carry significant implications. They suggest that interfaces can rescue derivations that narrow syntax alone cannot save and add a novel strategy to the inventory of solutions for the symmetry problem of UG: morphological repair via iconic form. Furthermore, this analysis opens a new perspective on areal linguistics. Similar patterns seem to also exist in languages across Southeast Asia (e.g., Thai, Burmese, Vietnamese), which suggests that the licensing head F may represent a parameterised grammatical strategy. Languages in this area appear to have converged on a shared morphological strategy to resolve symmetric merger, in contrast to the syntactic strategy (e.g., obligatory conjunctions) favored elsewhere. Ultimately, the AABB construction serves as a "compass", which reveals a deeply integrated architecture where syntax, phonology, and semantics cooperatively turn a potential computational crash into a meaningful expression.

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